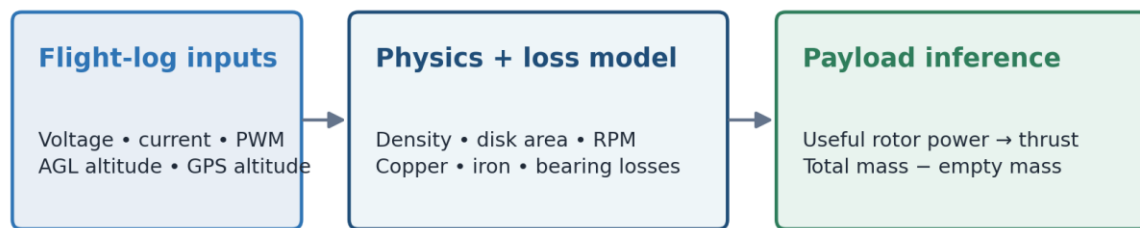


Payload Weight Estimation

A physics-informed model using flight-log power, motor command and altitude data

LOG-BASED PAYLOAD WEIGHT ESTIMATION



Model scope: from synchronized flight-log measurements to payload mass.

Prepared for General Aeronautics Pvt. Ltd.

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July 2026 | Final End-of-Internship Technical Report

Executive Summary

This project develops a payload-weight estimator for a six-rotor agricultural UAV using data already recorded by the flight controller. The final implementation isolates locally stable hover segments from ArduPilot BIN logs, removes the measured non-propulsive electrical baseline, converts motor commands to RPM, applies a calibrated motor-loss model and inverts actuator-disk theory to estimate total aircraft mass and payload.

The completed model uses an 18.75 kg aircraft tare mass, a 3.5 m local-altitude cutoff, an effective rotor area corrected for arm blockage, GPS-altitude air-density correction, a fixed propeller-efficiency coefficient of 0.718 and the final fitted loss coefficients $R_{\text{copper}} = 0.09667 \, \Omega$, $k_{\text{linear}} = 4.322131 \times 10^{-3} \, \text{W/RPM}$ and $k_e = 1.603907 \times 10^{-6} \, \text{W/RPM}^2$. Combining the previously separate linear-speed terms removed an exact identifiability problem in the regression.

Six field tests were conducted at nominal payloads of 0, 2, 4, 6, 8 and 10 kg, using a 30 s disarmed baseline followed by approximately two minutes of hover near 5 m AGL. Among the four clean loaded trials from 2 to 8 kg, three met the 5% payload-error target; the table-derived mean absolute percentage error was 5.89%, while the corresponding mean absolute total-mass error was 1.06%. The 10 kg flight was identified during testing as botched and is reported separately rather than treated as clean validation evidence.

Document status

The analytical model, final coefficients, corrected inference pipeline, field-test results, validation discussion, future-work roadmap and conclusion are complete. The report retains the final-day limitations and identifies the 10 kg trial as botched rather than presenting it as clean validation data.

The result demonstrates that the physics-informed method can recover total supported mass accurately from existing telemetry, while also showing that payload percentage error is amplified when a relatively small payload is obtained by subtracting the 18.75 kg tare mass. The remaining work is primarily improved coefficient characterization, repeated independent flights and stronger uncertainty and guard-condition handling.

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Report Scope

The internship also included optical-flow sensor validation and related automation work. Those activities are intentionally outside the technical scope of this report, except where the general lessons about log synchronization and validation discipline directly informed the payload-weight pipeline.

1. Introduction and Problem Statement

1.1 Problem Statement

According to actuator-disk (momentum) theory, the ideal power required by a rotor in hover is related to the thrust it produces by:

$$P_i = \frac{T^{3/2}}{\sqrt{2\rho A}}$$

Equation 1. Ideal induced power in hover.

where T is rotor thrust, ρ is air density, and A is the rotor disk or prop-wash area. Since hover thrust equals aircraft weight under steady conditions, this relationship provides a physical route from measured power back to total aircraft mass.

The ideal equation alone is insufficient for a real aircraft. Logged electrical power includes resistive heating, magnetic core losses, bearing friction, propeller inefficiency, onboard-electronics consumption and transient flight-control corrections. Environmental factors such as density, wind and ground effect also influence the power required for the same payload. The engineering problem is therefore to isolate a representative hover condition, account for the dominant non-ideal terms, and invert the corrected power balance without introducing more uncertain parameters than the data can support.

1.2 Project Objective

The objective is to estimate payload mass from an ArduPilot flight log using only signals already available on the hexacopter:

- BAT - voltage and current for total electrical input power;
- RCOU - all motor's output PWM for estimating the mean motor operating RPM;
- CTUN - rangefinder altitude for take-off baseline and stable-hover detection; and
- GPS - altitude above sea level for air-density calculation.

The estimator is intended as a post-processing and validation tool first. It would be integrated into the hexacopter's codebase (for on-screen output of the estimated payload post take-off) once rigorous testing is done and any and all necessary changes are made to the estimator.

1.3 Estimation Philosophy

The chosen approach combines first-principles structure with a small set of calibrated coefficients. This prevents physically impossible extrapolation while still allowing the model to represent a real motor-propeller system. The governing principle is:

$$P_e = P_{\text{loss}} + \frac{1}{\eta_p} \frac{T^{3/2}}{\sqrt{2\rho A_{\text{eff}}}}$$

Equation 2. Per-motor electrical power balance.

The useful portion of the measured electrical input is converted into thrust through actuator-disk theory. The final output is total supported mass across six rotors minus the known empty-aircraft mass.

1.4 Primary Assumptions

Assumption	Engineering interpretation
Steady hover	Vertical acceleration and mean climb rate are negligible within the selected window.
Approximately equal rotor loading	The six motors share the total vertical thrust sufficiently evenly that an average PWM and per-motor power allocation are representative.
Representative medians	Short spikes from control action or measurement noise do not dominate the selected median values.
Valid calibration domain	The manufacturer's PWM-to-RPM relationship and fitted loss coefficients remain applicable to the installed motor-propeller system.
No dominant gust event	The locally stable-altitude filter removes the strongest disturbances; wind is not explicitly estimated in the final code since its effects in power consumption increase is negligible. More on this below.
Positive useful power	Measured per-motor power must exceed the calculated loss power.

2. Factors Affecting Power Consumption

Seven real-world factors were investigated before fixing the final model architecture. Each was either represented directly, captured implicitly by existing telemetry, or excluded only after an order-of-magnitude assessment.

2.1 Air Density

Air density is governed mainly by pressure and temperature and therefore changes with altitude and weather. Lower density requires a higher induced velocity and more power for the same thrust. The final pipeline calculates density for every flight from the median GPS altitude during stable hover using a tropospheric lapse-rate model. The implementation uses the Indian Reference Atmosphere (IRA) instead of the ISA, thus assuming warmer sea level temperatures and lower sea level density.

2.2 Wind Conditions

A steady horizontal wind requires the aircraft to tilt so that the thrust vector balances both weight and aerodynamic drag. The development calculation used the standard drag equation and the vector magnitude of vertical and horizontal force:

$$D = \frac{1}{2} \rho C_D A_{\text{proj}} V_w^2, \quad T_{\text{net}} = \sqrt{(mg)^2 + D^2}$$

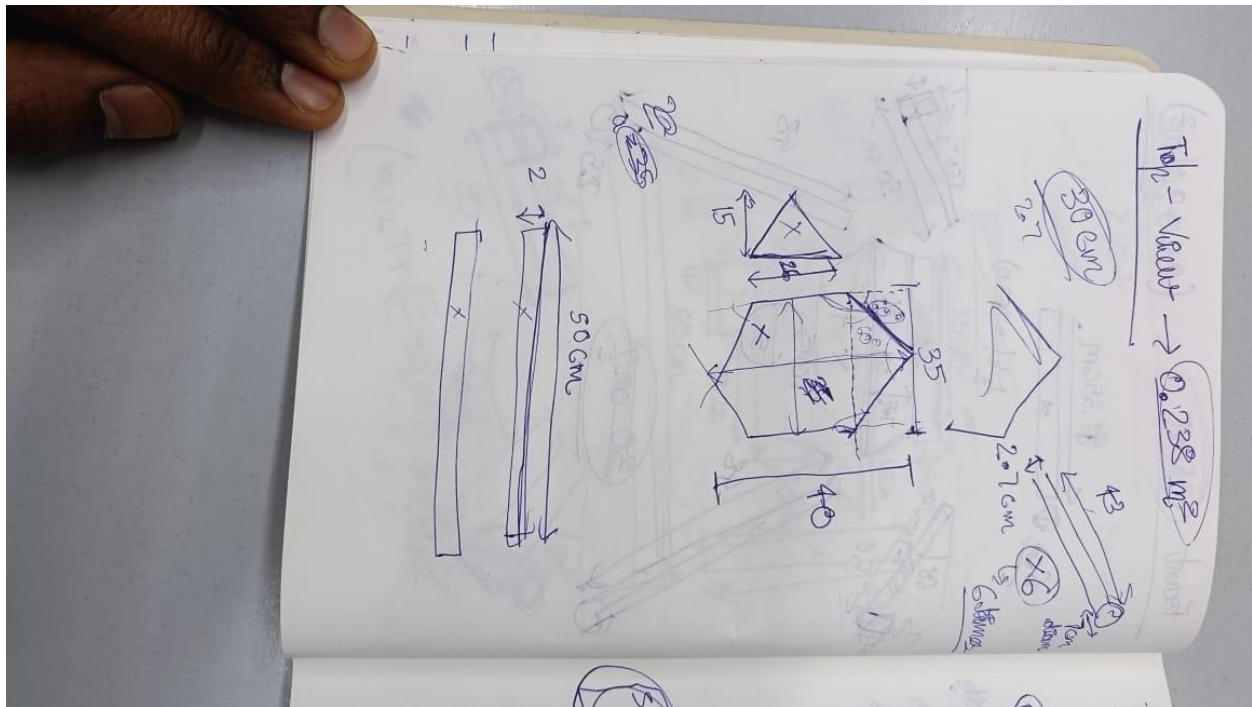
Equation 3. Steady wind drag and resultant thrust.

Assuming the wind has a horizontal wind angle of about 56.48° , the projected area is maximized to around 0.1738 m^2 . A conservative drag coefficient near 1.26, and a 10 m/s wind case (maximum possible wind speed within the operating envelope), the development study estimated roughly 13–14 N of drag, a 2–3° trim angle and about a 0.3% power increase at a 5 kg payload. This was below the expected overall estimator uncertainty, so wind was excluded from the implemented inference equation. Gust response remains different from steady wind because it depends on controller dynamics and airframe inertia.

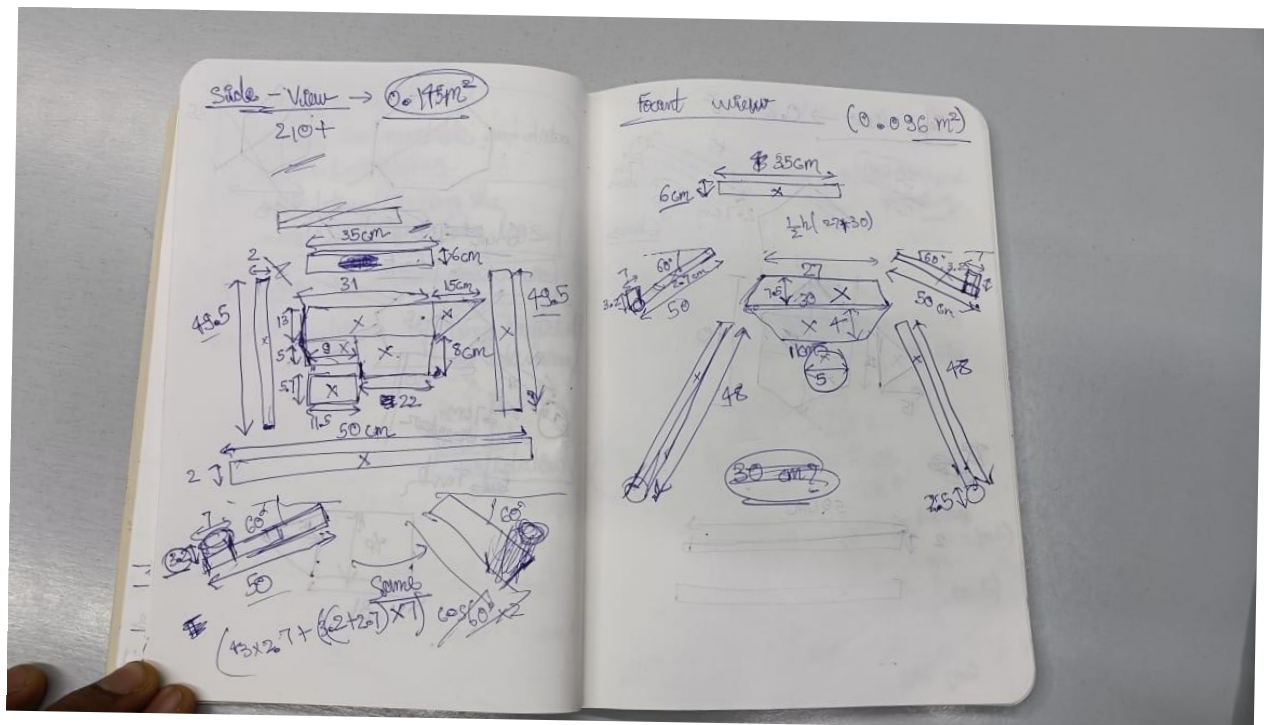
Here, the assumption is that the wind is entirely horizontal and has no vertical component.

Projected area calculations have been attached for the three side views.

For Coefficient of Drag calculations, skin friction has been neglected assuming the hexacopter is a bluff body. Drag area was calculated for components and divided by projected area – Front view was used for calculation of drag area and total projected area as well. I've attached my front, side and top view drawings for reference.



Top view projection



Side and Front view projections



 Cylindrical $\rightarrow 1.2 = G$

 Box $\rightarrow 1.05$

 Center Body $\rightarrow 1.1$

$A_f \cos \psi + A_s \sin \psi$

 Drag Area $\rightarrow$

 Σ Areas

Drag coefficient

Go front view that the volume!

$\rightarrow 1.15 \times 35 \times 6$ (Hexagonal base)

 $+ 0.75 \times 295.75$ (Tank)

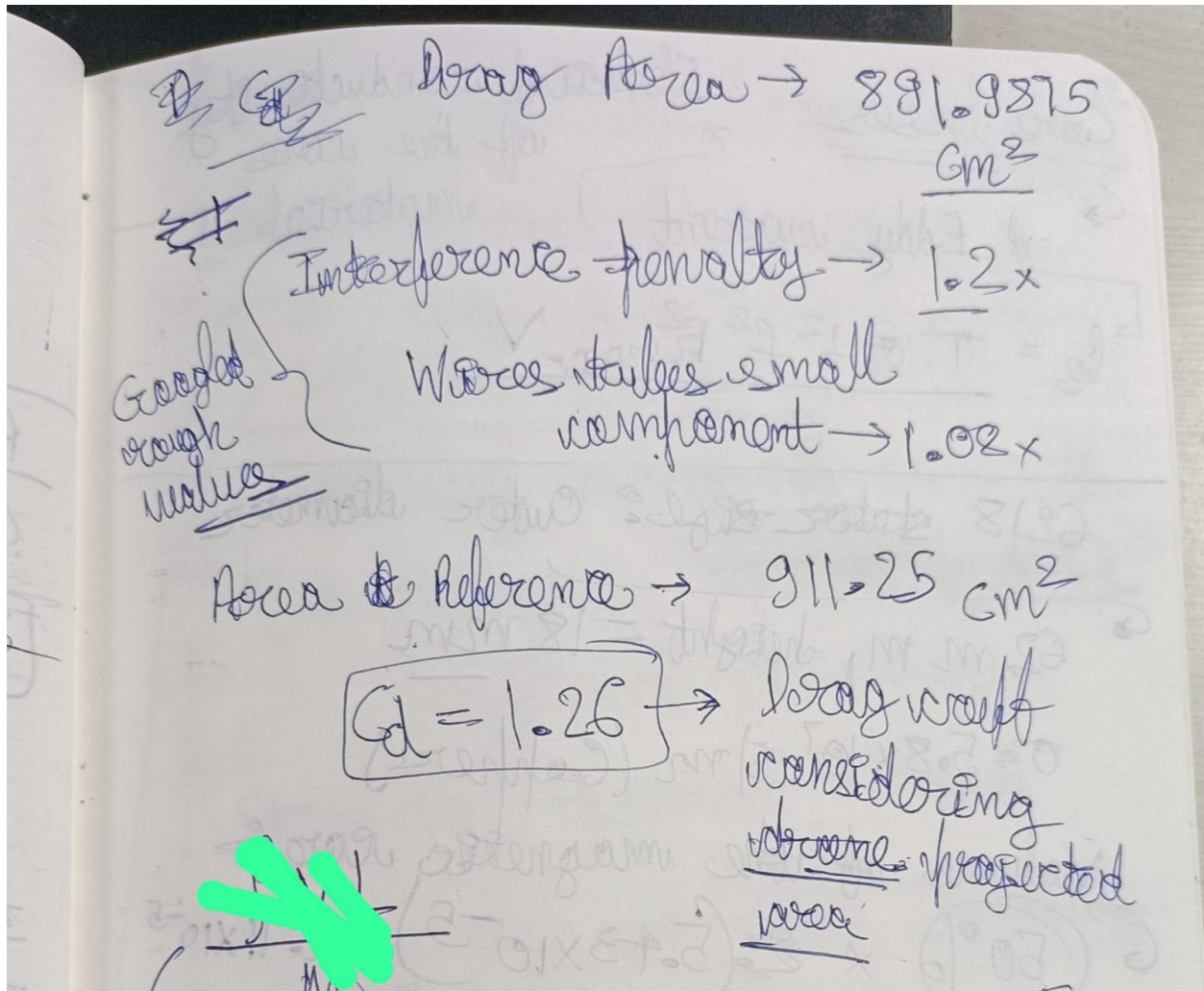
 $+ 1.2 \times 2 \times (48 \times 2)$ (Landing Tube)

 $+ 0.8 \times 2 \times (2.7 \times 50 \cos 60^\circ)$ (Retro

 $+ 1.15 \times (\pi 15^2)$ (Cylindrical Flow meter)

 arm

 tubes



Drag coefficient calculations are listed above.

Assumptions made – 1. Interference increases the drag coefficient by a factor of 1.2 (based on internet research and look-up on interference drag penalties for drones of such class)

1.08 - The additional drag penalty factor for wires, tubes and other components which haven't been considered in the simplified projected view.

2.3 Motor Thermal Efficiency

Winding resistance rises as copper temperature increases, raising I^2R loss. A temperature-correction framework was considered, including the approximately linear copper-resistivity trend over the expected operating range. It is not applied in the current pipeline because the installed system does not provide a reliable motor-winding temperature signal to the flight controller.

2.4 Drone Age and Wear

Bearing wear, motor degradation and damaged or aged propellers alter the fitted loss curve and propulsive efficiency. Rather than adding an unobservable age variable, the implemented architecture

supports periodic recalibration from a standardized hover or thrust test. The effective resistance, combined linear-speed term, quadratic-speed term and propeller-efficiency factor are therefore treated as configuration-dependent quantities that should be rechecked after powertrain or airframe changes.

2.5 Thrust Blockage

Part of each rotor disk is blocked by the arm-tube geometry. The blocked area is removed from the ideal disk area so the actuator model uses only the effective flow area:

$$A_{\text{eff}} = \pi r^2 - A_{\text{lost}}$$

Equation 4. Effective disk area.

For a 0.3048 m rotor radius, the geometric disk area is 0.29186 m². Subtracting 0.0081 m² (subtracting the projected area of each arm) gives an effective area of 0.28376 m² per motor, equivalent to a 2.78% area reduction.

2.6 Battery State of Charge

As state of charge falls, battery terminal voltage sags and current rises for a given demanded electrical power. The pipeline does not require a separate state-of-charge correction because it uses the instantaneous product of measured voltage and current. Battery electrical behavior is therefore included in the input power measurement, although any temperature-dependent motor-efficiency change caused by the higher current is represented only through the calibrated loss model accounting for increased coil resistance due to heating.

2.7 Ground Effect

Near the ground, reflected rotor downwash reduces the induced power required for a given thrust. The Cheeseman–Bennett ground-effect relation was evaluated during model development. The estimated correction at 2 m AGL was approximately 0.22%, while the significant region was concentrated below approximately one rotor diameter. The final code therefore rejects low-altitude data and performs inference only when SAlt (Rangefinder detected altitude) exceeds 3.5 m.

Implementation reconciliation

Earlier design notes used 2.0 m as the ground-effect exclusion threshold. The attached final pipeline implements 3.5 m. This report documents 3.5 m as the current model value and keeps the earlier 2.0 m figure only as development history.

2.8 Factor Disposition Summary

Factor	Treatment	Reason / implementation
Air density	Explicit	Calculated from stable-hover GPS altitude
Steady wind	Excluded after estimate	Calculated contribution $\approx 0.3\%$ in the defined worst case
Motor temperature	Documented limitation	No winding-temperature telemetry
Age and wear	Periodic calibration	Refit sensitive loss/efficiency coefficients
Thrust blockage	Explicit	0.0081 m^2 removed per rotor
Battery state of charge	Implicit	Captured through measured $V \times I$
Ground effect	Filtered	Stable-hover data must be above 3.5 m

3. Current Payload-Weight Estimation Model

3.1 Actuator-Disk Derivation

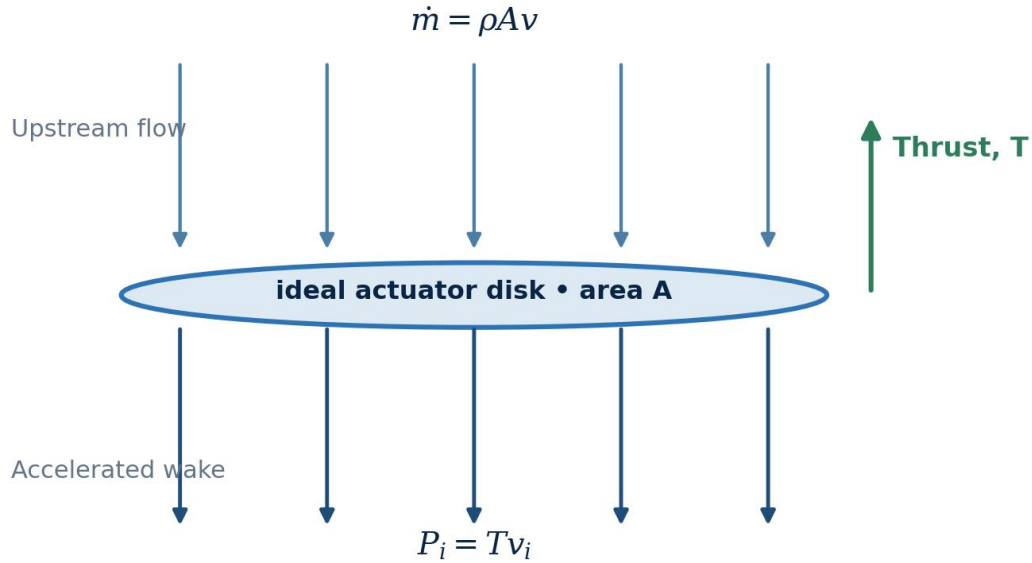


Figure 1. Ideal actuator-disk representation used for the thrust–power relation.

For an ideal rotor in hover, the induced velocity at the disk is v_i and the far-wake velocity is $2v_i$. The mass-flow rate through the disk is approximately $\rho A v_i$. Applying momentum conservation gives $T = 2\rho A v_i^2$. The useful, induced power is $T v_i$. Eliminating v_i produces Equation 1, $P_i = T^{3/2}/\sqrt{2\rho A}$.

The ideal disk assumes uniform induced velocity and no tip, swirl, profile-drag or non-uniform inflow losses. A real two-bladed propeller therefore requires more shaft power than the ideal induced power. The model represents this through the propeller efficiency coefficient η_p .

3.2 Electrical Power Balance

For one rotor-powertrain branch, the measured electrical power is represented as the sum of modeled motor losses and the electrical power required to supply the useful induced power:

$$P_e = P_{\text{loss}} + \frac{1}{\eta_p} \frac{T^{3/2}}{\sqrt{2\rho A_{\text{eff}}}}$$

Equation 5. Final per-motor electrical-power model.

In the attached code, per-motor electrical power P_e is calculated as the median total battery power (baseline power consumption was subtracted) during stable hover divided by six. This assumes that the logged total power can be allocated evenly across the six motors. The baseline non-propulsive current is

also removed when estimating per-motor current for copper loss; this convention must be retained when interpreting the final test results.

3.3 Motor and Drivetrain Loss Model

$$P_{loss} = R_{copper} I_m^2 + k_{linear} N + k_e N^2$$

Equation 6. Implemented loss model.

The final loss model contains three fitted terms: $R_{copper} I_m^2$ for effective current-squared loss, $k_{linear} N$ for combined linear-speed loss and $k_e N^2$ for quadratic-speed loss. The earlier $k_h N$ and k_{bw} terms were exactly collinear because $\omega = 2\pi N/60$, so they were merged into k_{linear} . These are effective calibration parameters tied to the manufacturer curve and installed powertrain, not independently measured material properties.

Avoiding double counting

The separate aerodynamic windage term remains excluded because fixed propeller efficiency already represents major propeller non-idealities. The two linear-speed loss terms were merged because RPM and angular speed are deterministically related.

3.4 Air-Density Model

$$\rho(h) = \rho_0 \left(1 - \frac{Lh}{T_0} \right)^{\left(\frac{g}{RL} - 1 \right)}$$

Equation 7. Implemented tropospheric density relation.

The stable-hover median GPS altitude h is inserted into Equation 7. The constants are $R = 287.058$ J/(kg·K), $L = 0.0065$ K/m, $g = 9.81$ m/s², $T_0 = 303.15$ K and $\rho_0 = 1.1551$ kg/m³. The exponent in the final code is 4.25758. This is a dry-air, fixed-reference-temperature model; humidity and flight-specific ambient temperature are not independently measured in this version of density calculation.

3.5 PWM-to-RPM Conversion

$$u_{\%} = 100 \frac{\overline{PWM} - PWM_{min}}{PWM_{max} - PWM_{min}}, \quad N = \text{interp}(u_{\%})$$

Equation 8. PWM normalization and tabulated RPM interpolation.

The arithmetic mean of motor outputs C1–C6 is converted to a throttle percentage using the documented 1050–1950 μ s operating pulse range. RPM is then obtained by one-dimensional

interpolation through the manufacturer load-performance table. The final lookup contains 23 points from 40%/2455 RPM to 100%/5442 RPM. Values outside the table are clipped to the nearest endpoint by NumPy interpolation.

3.6 Model Inversion

Rearranging the electrical power balance gives the thrust supported by one rotor:

$$T = \left[(P_e - P_{\text{loss}}) \eta_p \sqrt{2\rho A_{\text{eff}}} \right]^{2/3}$$

Equation 9. Per-rotor thrust inferred from useful electrical power.

For six nominally equal rotors, payload mass is then:

$$m_{\text{payload}} = \frac{n_r T}{g} - m_{\text{empty}}$$

Equation 10. Payload-mass calculation.

The final implementation uses $n_r = 6$, $g = 9.81 \text{ m/s}^2$ and an empty-aircraft mass of 18.75 kg. If corrected per-motor input power minus modeled loss is not positive, the log is rejected because the inversion is physically undefined.

3.7 Symbol and Unit Definition

Symbol	Definition	Unit
P_e	Per-motor electrical input power	W
P_i	Ideal induced power delivered to the air	W
P_{loss}	Modeled current-squared, linear-speed and quadratic-speed losses	W
T	Thrust supported by one rotor	N
ρ	Air density at stable-hover GPS altitude	kg/m ³
A_{eff}	Effective rotor disk area after blockage	m ²
η_p	Calibrated propeller / aerodynamic efficiency	dimensionless
I_m	Estimated current per motor after baseline-current removal	A
N	Motor rotational speed	RPM
m_{empty}	Aircraft mass without payload	kg
m_{payload}	Estimated payload mass	kg

4. Model Parameters and Justification

The parameter set contains four distinct classes: universal physical constants, measured platform geometry, manufacturer-derived operating data, and calibrated or engineering-estimate coefficients. Keeping these categories separate is essential because they have different uncertainty and recalibration requirements.

4.1 Platform and Geometry Parameters

Parameter	Value	Unit	Source / method	Justification
Number of rotors, n_r	6	count	Aircraft configuration	Used to scale per-rotor supported mass to total aircraft mass.
Empty mass, m_{empty}	18.75	kg	Flight-test-team configuration update	Includes approximately 0.25 kg of additional installed components that were absent from the earlier 18.5 kg estimate.
Rotor radius, r	0.3048	m	24-inch propeller geometry	Matches the 24×8 propeller family used for the performance lookup.
Blocked area, A_{lost}	0.0081	m ² /rotor	Arm-tube geometry	Represents the portion of disk area obstructed by structure.
Effective area, A_{eff}	0.2837635	m ² /rotor	$\pi r^2 - A_{\text{lost}}$	A 2.78% reduction relative to the geometric disk.
PWM minimum	1050	μs	Manufacturer specification	Lower bound of the documented operating pulse range.
PWM maximum	1950	μs	Manufacturer specification	Upper bound of the documented operating pulse range.

4.2 Atmospheric Constants

Parameter	Value	Unit	Reason for value
R	287.058	J/(kg·K)	Specific gas constant for dry air
L	0.0065	K/m	Tropospheric temperature lapse-rate approximation
g	9.81	m/s ²	Gravity used in both atmosphere and mass conversion
T_o	303.15	K	30 °C IRA reference selected for the operating environment
ρ_o	1.1551	kg/m ³	Reference density paired with other IRA baseline parameters
Exponent	4.25758	—	Computed as $g/(RL) - 1$

4.3 Loss and Efficiency Coefficients

Coefficient	Final value	Unit	Determination	Model role
R_copper	0.09667	Ω	Bounded curve fit to the manufacturer power/current/RPM/thrust table	Effective current-squared electrical-loss coefficient
k_linear	4.322131×10^{-3}	W/RPM	Combined fitted linear-speed coefficient after merging k_hN and k_bw	Identifiable linear-speed loss term
k_e	1.603907×10^{-6}	W/RPM ²	Bounded curve fit with η_{prop} fixed	Quadratic-speed loss surrogate
η_{prop}	0.718	—	Fixed at a realistic manufacturer-data-based propeller-efficiency value	Converts corrected electrical power after modeled losses to induced power

These are implementation coefficients tied to the manufacturer test curve, the selected 0.718 propeller-efficiency factor and the installed motor-propeller-ESC configuration. They should be recalibrated after hardware or airframe changes and should not be interpreted as independently measured material constants.

4.4 Signal-Processing and Stability Parameters

Parameter	Implemented value	Justification
Timestamp tolerance, TOL	60,000 μ s	Nearest-sample allowance used by merge_asof; accommodates the lowest-rate GPS stream while preserving local alignment.
Rolling stability window	2.5 s	Long enough to reject short altitude corrections while remaining local within a hover segment.
Altitude cutoff	SAlt > 3.5 m	Excludes take-off, landing and the previously quantified ground-effect region; final tests hovered near 5 m AGL.
Stability threshold	Flight-specific 50th percentile of rolling SAlt standard deviation	Adaptive to the noise level of each flight rather than using one fixed sensor-noise value.
Minimum rolling count	20 rows	Prevents a rolling statistic based on only a few samples.
Ground baseline	SAlt < 0.05 m	Estimates non-propulsive current while the aircraft is effectively at zero height.
Summary statistic	Median	Reduces sensitivity to transient current, power, PWM and altitude spikes.

4.5 Parameter Confidence Classification

Confidence class	Parameters	Interpretation
High	g , R , rotor count, PWM endpoints, RPM lookup points	Defined physical constant, configuration value or manufacturer table.
Medium	empty mass, rotor radius, blockage area, 3.5 m cutoff	Measurable on the aircraft but must match the exact test configuration.
Calibration-dependent	R_{copper} , k_{linear} , k_e , η_{prop}	Must be checked when the motor, propeller, ESC, airframe condition or calibration convention changes.
Engineering estimate	ρ_0 , T_0 , 60 ms tolerance, 2.5 s window	Reasoned values that require sensitivity checks and independent flight validation.

5. Methodology Used to Determine Parameter Values

5.1 Development Data Sources

Parameter determination used three complementary sources: (1) manufacturer load-performance data relating throttle, current, power, thrust and RPM; (2) aircraft flight logs containing voltage, current, motor output and altitude; and (3) first-principles or standard engineering equations used to constrain poorly observable loss terms. This prevented the regression from assigning every unexplained watt to a single coefficient.

5.2 Stable-Hover Data Preparation

1. Power was calculated as voltage multiplied by current at each aligned timestamp.
2. CTUN SAlt was selected as the local height signal because earlier barometer-based extraction was too noisy for hover-window selection.
3. Altitude and power traces were plotted together to identify rising, descending and approximately level portions of the flight.
4. Linear trend checks were used during early manual validation; datasets with strong altitude or power drift were rejected.
5. The final implementation replaces most manual trimming with the 2.5 s rolling local-stability rule.

5.3 Rejection of Purely Empirical Polynomial Models

A second-order power-versus-weight fit and a $T^{3/2}$ physics fit both described the manufacturer data well over the measured range. Higher odd-order polynomials were rejected because they could turn downward outside the fitted range and imply that required power decreases as weight rises. Even a high R^2 inside the calibration range would not prevent this non-physical extrapolation. Actuator-disk theory was therefore selected as the model backbone.

5.4 Separation of Useful Power from Losses

An early attempt used the datasheet's g/W efficiency directly. That metric was rejected because it combines aerodynamic and electrical behavior into a single output ratio. It cannot distinguish useful induced power from copper, iron, bearing and propeller losses. The final method explicitly separates the power balance before solving for thrust.

5.5 Copper-Resistance Identification

Initially, the full residual between electrical power and ideal induced power was divided by I^2 . The apparent resistance varied strongly with operating point because core, mechanical and aerodynamic effects were being absorbed into one term. The final coefficient was therefore obtained only as part of a

bounded multivariable fit after fixing η_{prop} at 0.718 and expressing the remaining loss curve using I^2 , RPM and RPM^2 . The resulting effective current-squared coefficient was $R_{\text{copper}} = 0.09667 \, \Omega$.

A handheld multimeter reading was not used as the primary estimate because the meter resolution and lead/contact resistance were too large relative to the low winding resistance. This is an example of rejecting a measurement method whose uncertainty exceeded the quantity being measured.

5.6 Combined Linear and Quadratic-Speed Coefficients

The earlier formulation used separate hysteresis and bearing-friction terms proportional to RPM and angular speed. Because angular speed is a fixed multiple of RPM, those terms form the same regression column and cannot be independently identified. They were merged into a single coefficient, k_{linear} . With η_{prop} fixed at 0.718 and the manufacturer power/current/RPM/thrust table used as the fitting dataset, the final bounded curve fit produced $k_{\text{linear}} = 4.322131 \times 10^{-3} \, \text{W/RPM}$ and $k_e = 1.603907 \times 10^{-6} \, \text{W/RPM}^2$.

The fitted manufacturer-curve reconstruction gives $R^2 \approx 0.99979$, a mean absolute power residual of approximately 6.06 W and a maximum residual of approximately 21.8 W across the 23 tabulated operating points. This is strong in-sample agreement, but the coefficient values remain correlated and are therefore treated as effective calibration parameters rather than uniquely identified physical loss constants.

5.7 Propeller Efficiency and Final Fit Convention

Propeller efficiency was fixed at 0.718 before the final loss-curve fit. Locking this realistic aerodynamic factor reduced the number of free variables and prevented the useful-power scale from shifting freely against the loss coefficients. The separate ω^3 propeller-windage term remained excluded to avoid double counting non-ideal propeller behavior already represented by η_{prop} .

This final convention supersedes the earlier 0.7185656 implementation and the separate k_h and k_b linear-speed terms. In the completed pipeline, all fitted residual loss is represented only by $R_{\text{copper}} I_{\text{m}}^2$, $k_{\text{linear}} N$ and $k_e N^2$.

5.8 PWM-to-RPM Map

The ESC accepts a PWM command but regulates the motor under load; RPM cannot be estimated reliably as KV multiplied by voltage. The final method therefore uses the manufacturer's measured load-performance RPM table. PWM is first normalized to throttle percentage, after which linear interpolation gives RPM at the median hover command.

5.9 Environmental and Geometry Parameters

Air-density constants were selected for IRA reference and applied to GPS altitude through the tropospheric lapse-rate relation. The rotor radius follows the 24-inch propeller size. Blocked area was calculated from the arm-tube geometry and directly subtracted from the disk. Wind and ground effect were quantified separately and then excluded by engineering significance and data filtering, respectively.

5.10 Recalibration Methodology

Use the complete aircraft configuration and measure the empty mass directly with all installed components.

Record several stable-hover or controlled-thrust operating points across the payload and RPM range, including a verified motors-disarmed electronics baseline.

Extract synchronized voltage, current, PWM, altitude and environmental data using the same inference pipeline.

Fix the rotor geometry, atmosphere convention, PWM-to-RPM lookup and propeller-efficiency convention before fitting losses.

Fit only the identifiable coefficients R_{copper} , k_{linear} and k_e within physically plausible bounds; report covariance or sensitivity rather than treating the terms as exact material properties.

Validate on independent flights not used for fitting and compare payload MAPE, absolute error, signed bias, repeatability and total-mass error.

Validate on separate logs not used in the fit and compare payload error, bias and repeatability.

6. Final Testing and Inference Pipeline

6.1 Input Files and Log Messages

The final Python script processes every lowercase .bin file in a configured Windows test folder. Each file is opened through pymavlink and read message by message. Four message families are retained:

Message	Fields	Pipeline purpose
GPS	TimeUS, Alt	Altitude used for air-density correction
CTUN	TimeUS, SAlt	Rangefinder altitude used for baseline and stable-hover detection
RCOU	TimeUS, C1–C6	Six motor-output pulse widths; arithmetic mean becomes PWM_avg
BAT	TimeUS, Volt, Curr	Instantaneous battery voltage and current; Curr is used rather than cumulative CurrTot

6.2 Timestamp Alignment

The message streams are asynchronous and have different sample rates. CTUN is used as the left-hand timeline. RCOU, BAT and GPS are successively joined to the nearest CTUN row with a 60 ms tolerance:

$$|t_{\text{source}} - t_{\text{CTUN}}| \leq 60 \text{ ms}$$

Equation 11. Nearest-timestamp acceptance condition.

Rows without a valid nearby sample are dropped. The resulting aligned data are saved as one CSV per BIN file, providing a traceable intermediate artifact for plotting and manual audit.

6.3 Power and Motor-Command Features

At each RCOU timestamp, PWM_avg is the arithmetic mean of C1 through C6. Total electrical power is Volt×Curr.

6.4 Baseline Current and Electronics-Power Correction

The disarmed near-ground interval is used to estimate the non-propulsive electronics baseline. The final revision removes this baseline from both the current used in the fitted loss model and the total electrical power supplied to the six propulsion branches. This correction was added after the first field-test pass showed that leaving auxiliary power inside P_e caused systematic payload overestimation.

$$I_m = (\text{median current in stable hover} - \text{median current at } SAlt < 0.05 \text{ m}) / 6$$

The field-test procedure deliberately retained approximately 30 s with the aircraft disarmed before take-off so the baseline could be estimated. The corrected per-motor propulsion input is then obtained

only after removing the measured auxiliary contribution; this convention must remain unchanged when the coefficients or results are reproduced.

6.5 Local Stable-Hover Logic

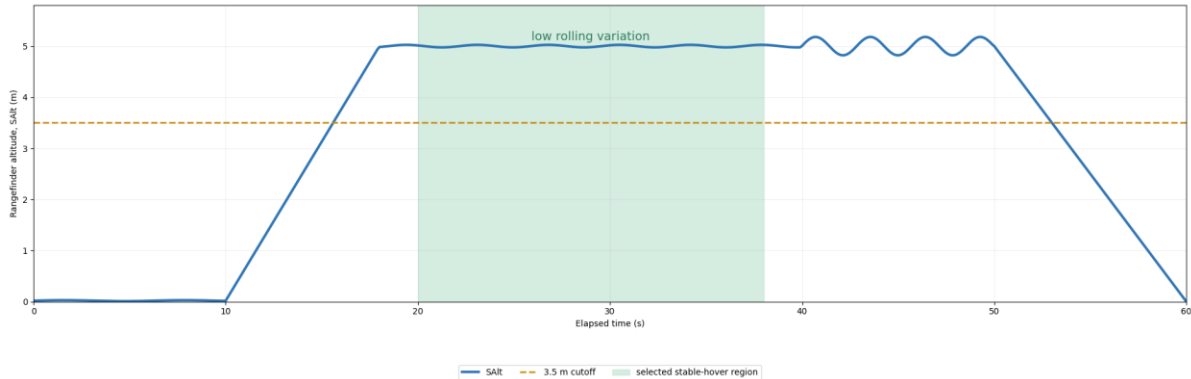


Figure 2. Conceptual illustration of the implemented local stable-hover selection.

TimeUS is converted to a pandas timedelta index and SAlt rolling statistics are calculated over the trailing 2.5 s. A row is classified as stable hover only when SAlt exceeds 3.5 m, the rolling standard deviation is below the flight-specific 50th percentile, and the rolling window contains at least 20 samples. The final flight campaign hovered near 5 m AGL, leaving margin above the implemented cutoff.

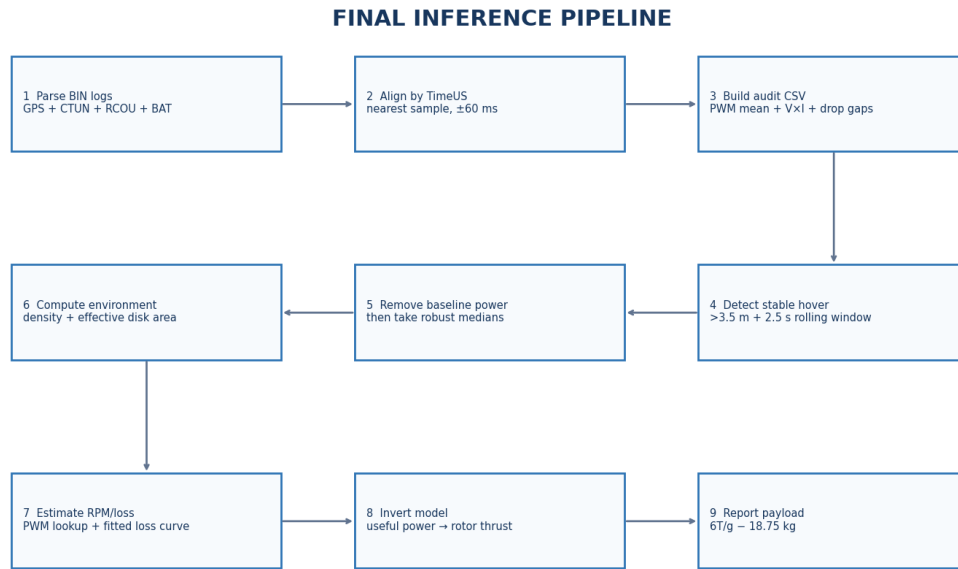
6. SAlt is greater than 3.5 m;
7. the 2.5 s rolling standard deviation is below the 50th percentile of rolling standard deviation for that flight; and
8. the rolling window contains at least 20 samples.

The percentile threshold adapts to each log's noise level and removes the most variable half of eligible rolling windows. It is a relative stability test: if an entire flight is poor, it may still identify the least unstable portion. Final validation should therefore interpret the selected-window plots together with the numerical output.

6.6 Robust Representative Values

The pipeline reduces the selected stable-hover rows to four medians: GPS altitude, mean motor PWM, total electrical power and battery current. Median aggregation was chosen over mean aggregation for these flight-level summaries because it is less sensitive to isolated power spikes or controller corrections.

6.7 Computation and Output Sequence



Guard condition: corrected useful power must remain positive

Figure 3. End-to-end execution logic in the completed testing pipeline, including the 3.5 m hover cutoff, electronics-baseline power correction and 18.75 kg tare mass.

For each usable log, the script computes density, maps PWM to RPM, converts RPM to rad/s, evaluates Equation 6, checks the positive-useful-power guard, evaluates Equations 9 and 10, and prints the payload estimate to four decimal places.

6.8 Current Failure and Skip Conditions

Condition	Current behavior
Missing GPS, CTUN, RCOU or BAT stream	File is skipped with a message.
No aligned sample within ± 60 ms	Unmatched rows are removed by dropna.
Calculated useful power ≤ 0	File is skipped; hover window or calibration must be checked.
No near-ground baseline samples	Current baseline becomes NaN; the current code has no dedicated error message.
No stable-hover rows	Representative medians become undefined; the current code has no explicit empty-window guard.
PWM outside lookup domain	NumPy interpolation clips to 40% or 100% endpoint RPM.

6.9 Reproducibility and Audit Trail

The pipeline recreates the `aligned_csv` output folder for each run and writes one synchronized CSV per processed BIN file. This makes the exact rows used by downstream calculations inspectable. Because the folder is deleted and rebuilt at start-up, any manually added files inside `aligned_csv` must be stored elsewhere before execution.

7. Development, Bug Fixes and Validation

This section records the evolution of the estimator separately from the final model description. It includes both technical corrections and validation activities completed before the final flight-test campaign.

7.1 Development Evolution

Stage	Problem / observation	Resolution
Initial extraction	Parsed current and voltage and calculated power, then discovered the first export/filter method was incorrect.	Re-exported using message filtering and verified values against plots.
Manual hover filtering	Removed climb and descent portions by comparing power with altitude.	Established the need for an automated local-stability rule.
Altitude signal choice	Barometer-based height was too noisy for the selection task.	Adopted CTUN SAlt as the local height signal.
Curve-fitting exploration	Polynomial and $T^{3/2}$ relationships were compared.	Selected the physics relation to prevent non-physical extrapolation.
Efficiency metric	Tried to use datasheet g/W efficiency as the correction.	Rejected it because it blends useful power and multiple loss mechanisms.
Loss identification	Attributed the full residual to I^2R , producing implausibly high resistance.	Separated copper, core and mechanical losses through constrained multivariable modeling.
Motor speed	Considered KV×voltage and linear voltage assumptions.	Used the measured manufacturer throttle-to-RPM lookup under load.
Environmental terms	Investigated density, wind, ground effect, blockage, SoC and aging.	Included only terms with sufficient significance and observability.
Automation	Manual time alignment and visual trimming were slow and subjective.	Implemented TimeUS nearest joins, rolling stability, medians and batch output.
Final field-test calibration	Initial 0–10 kg tests overestimated payload and exposed a tare-mass and baseline-power mismatch.	Updated tare mass to 18.75 kg, removed electronics baseline power and refit identifiable loss coefficients with η_{prop} fixed.
Coefficient identifiability	Separate k_{hN} and k_{bw} terms use the same independent variable.	Merged them into $k_{linear N}$ and retained one quadratic-speed term.

7.2 Bug Fixes and Model Corrections

Issue	Root cause / symptom	Correction
Incorrect data extraction / filtering	Power trends did not match plotted flight behavior.	Corrected the export and checked extracted data against altitude and power plots.
Unusable flight dataset	Strong drift and poor hover stability made a dataset unsuitable.	Rejected the dataset rather than forcing it into the model.
Mis-indexed manufacturer operating point	An incorrect power-throttle pairing caused weight overestimation.	Corrected the selected manufacturer row and repeated the calculation.

Issue	Root cause / symptom	Correction
Inflated winding resistance	Residual/ I^2 treated iron and mechanical losses as copper loss.	Introduced speed-dependent loss terms and a multivariable fit.
Low-ohm multimeter uncertainty	Meter resolution and lead/contact resistance were comparable to the winding resistance.	Used model-based identification rather than the handheld reading.
Double-counted propeller losses	A separate ω^3 windage term overlapped with calibrated propeller efficiency.	Removed k_{ω^3} ; the later k_h and k_b linear terms were merged into k_{linear} .
Ambiguous loss notation	Development notes used k_w , k_h and k_b inconsistently while the model was evolving.	Final notation distinguishes removed aerodynamic windage k_w from the merged identifiable term k_{linear} .
PWM statistic comment mismatch	Code comment says median, implementation uses arithmetic mean of C1–C6.	Report and model description follow the executable <code>mean(axis=1)</code> .
Ground-effect cutoff drift	Earlier design values included 2.0 m and 1.5 m during pipeline development.	The edited final report and completed validation convention use <code>SAlt > 3.5 m</code> .
Test-condition note mismatch	A development note listed a different manufacturer test temperature.	The retained X6 Plus manufacturer sheet states 25 °C and sea-level test conditions.
Unaccounted added components	The 18.5 kg tare omitted approximately 0.25 kg of installed hardware.	Revised empty-aircraft mass to 18.75 kg.
Auxiliary power inside thrust balance	Baseline current was removed for loss current, but baseline electrical power initially remained in P_{total} .	Subtracted the motors-disarmed electronics baseline from total power before actuator-disk inversion.
Redundant linear coefficients	k_{hN} and $k_{b\omega}$ were exactly collinear.	Combined both terms into k_{linear} and refit R_{copper} , k_{linear} and k_e with $\eta_{\text{prop}} = 0.718$ fixed.

7.3 Validation Completed So Far

Validation layer	Method	Current status
Dimensional and monotonic checks	Confirmed that higher useful power produces higher thrust and that lower density increases required induced power.	Passed at equation level.
Manufacturer-table consistency	Compared model behavior with measured throttle, current, power, thrust and RPM operating points.	Used for coefficient fitting and row-by-row sanity checks.
Physics-fit quality	Reconstructed the 23-point manufacturer operating curve with the final three-term loss model and fixed η_{prop} .	$R^2 \approx 0.99979$; mean absolute residual ≈ 6.06 W; in-sample calibration evidence only.
Operating-point spot checks	Recovered thrust/weight from individual manufacturer rows after correcting row selection.	Development checks reported close agreement.
Ground-effect significance	Applied Cheeseman–Bennett estimate at operational height.	$\approx 0.22\%$ at 2 m; excluded through altitude filtering.
Wind significance	Used projected area, conservative C_D and 10 m/s wind.	$\approx 13\text{--}14$ N drag and $\approx 0.3\%$ power increase in the defined case; excluded.
Log synchronization	Merged asynchronous messages by TimeUS and exported aligned CSVs.	Pipeline produces inspectable combined records.
Hover-selection logic	Applied altitude cutoff, rolling standard deviation and minimum sample count.	Automated local selection implemented; final flight-window plots still required with results.

Validation layer	Method	Current status
Guard checks	Reject missing message families and non-positive useful power.	Implemented in the batch script.
Final payload flight matrix	Conducted nominal 0, 2, 4, 6, 8 and 10 kg flights after a 30 s disarmed baseline and near-5 m hover.	Three of four clean loaded trials from 2–8 kg were below 5% payload error; clean loaded MAPE = 5.89%.
Total-mass accuracy	Converted each final payload result back to supported aircraft mass using the 18.75 kg tare.	Mean absolute total-mass error = 1.06% for the clean loaded 2–8 kg trials.

Important interpretation

The final flight tests are a first validation campaign, not a complete uncertainty study. Only one flight was run at each nominal payload, the 10 kg flight was identified as botched, and the negative zero-payload result exposes remaining low-load sensitivity. Final claims therefore distinguish clean loaded trials, the botched trial and the diagnostic zero-payload case.

7.4 Current Code-Review Observations

9. The flight-specific percentile threshold is relative rather than absolute; it always ranks each flight's windows even when the complete flight is noisy.
10. The current test matrix contains only one flight per payload, so repeatability, confidence intervals and day-to-day robustness cannot yet be quantified.
11. Mean PWM and equal current allocation provide a tractable six-motor approximation but do not capture individual rotor trim or current imbalance.
12. The zero-payload result became negative after baseline-power correction, indicating sensitivity to baseline selection, tare mass and low-load coefficient transfer.
13. The batch output directory is rebuilt on every run; aligned CSV evidence must be archived separately for traceable validation.

8. Current Status and Deliverables

Work package	Status	Evidence / output
Problem formulation	Complete	Physics-based payload inference defined.
Factor assessment	Complete	Seven environmental/platform effects evaluated and dispositioned.
Final analytical model	Complete	Actuator-disk inversion plus copper, core, bearing, efficiency, density and blockage terms.
Parameter set	Complete for testing	Values, source classes and rationales documented.
BIN-log extraction	Complete	GPS, CTUN, RCOU and BAT parsed with pymavlink.
Timestamp alignment	Complete	Nearest TimeUS join with 60 ms tolerance.
Stable-hover logic	Complete	3.5 m cutoff, 2.5 s local rolling variation and minimum sample count.
Batch inference	Complete	One payload estimate and aligned CSV per usable log.
Development validation	Complete	Manufacturer-data fit and engineering significance checks documented.
Independent flight-test results	Complete – initial campaign	0–10 kg matrix documented in Section 9; 10 kg flight identified as botched.
Implementation roadmap	Complete	Coefficient, repeatability, baseline and software improvements documented in Section 10.

The principal technical deliverable is `testing_pipeline_final.py` together with this report. The script provides an auditable bridge from raw ArduPilot BIN logs to a payload estimate; the report records the assumptions required to interpret that estimate correctly.

9. Results and Discussion

Final field-test campaign

Six payload configurations were tested on 30 July 2026. Each run included approximately 30 s with the aircraft disarmed for baseline-power estimation, followed by roughly two minutes of hover near 5 m AGL. The 10 kg test was marked as botched during the campaign and is retained for transparency but separated from the clean validation subset. The "botched" test case for 10 kg is termed so because at around 1 minute hover time, the drone had switched to RTL mode and landed. I've still tried to filter the data and then use it for further calculations as listed below.

9.1 Test Matrix

Flight time	Known payload	Status	Interpretation
12:13:01	0 kg	Diagnostic	Negative inferred payload after final baseline correction
12:28:56	2 kg	Usable	Clean loaded trial
13:48:31	4 kg	Usable	Largest clean-trial payload error
13:55:38	6 kg	Usable	Clean loaded trial
14:00:42	8 kg	Usable	Clean loaded trial; slight under-estimate
14:22:12	10 kg	Botched	Reported separately; not treated as clean validation

All flights used the same model coefficients and processing pipeline. Because only one flight was conducted at each nominal payload, the campaign measures point accuracy but does not yet establish repeatability.

9.2 Estimated Payload and Error Metrics

Known kg	Estimated kg	Signed error kg	APE %	Loss W/motor	Corrected input W/motor
0	-0.9441	-0.9441	—	34.6037	316.4901
2	2.0982	+0.0982	4.91	40.1027	397.2313
4	4.5168	+0.5168	12.92	46.8277	467.9485
6	6.2172	+0.2172	3.62	52.5986	520.7175
8	7.8311	-0.1689	2.11	60.2426	574.4863
10*	10.7203	+0.7203	7.20	64.8769	665.1986

*The 10 kg flight was identified as botched and is not included in the clean-trial headline metric.

Accuracy summary

For the clean loaded 2, 4, 6 and 8 kg trials, payload MAPE was 5.89%; three of the four trials met the 5% target. Expressed as total supported mass rather than payload-after-tare subtraction, the same four flights produced a mean absolute percentage error of 1.06%. Including the botched 10 kg result gives a table-derived loaded-payload MAPE of 6.15%. The final console output recorded 6.4635%; the transparent table-derived metrics are used here because the earlier script aggregation used legacy error handling.

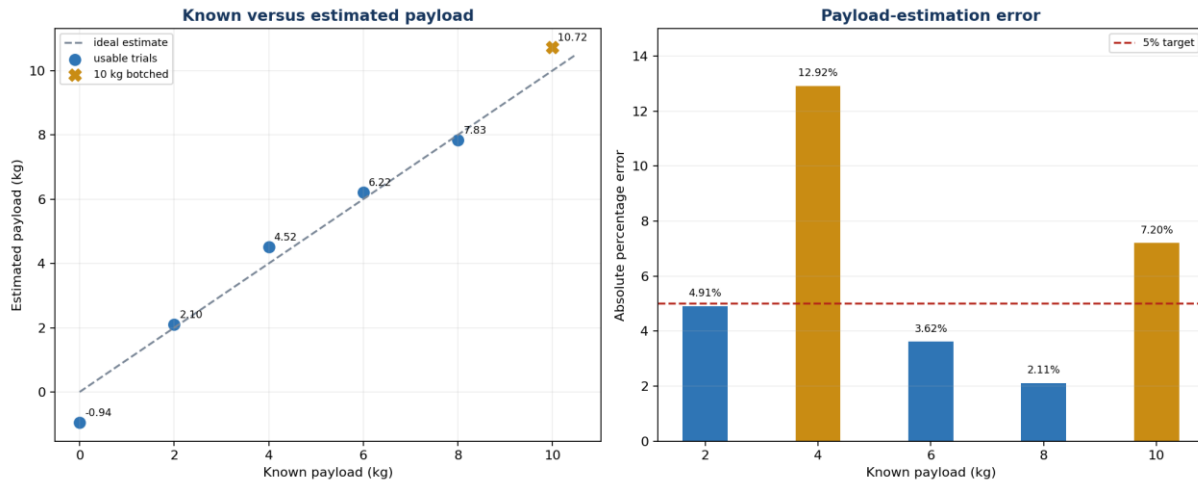


Figure 4. Final known-versus-estimated payload and absolute percentage error. The 10 kg trial is marked as botched.

9.3 Stable-Hover and Power Results

The representative values were medians over rows that satisfied the final 3.5 m altitude cutoff, 2.5 s rolling-stability rule and minimum sample count. The test procedure provided a motors-disarmed baseline before each flight, and the final correction removed that auxiliary power from the propulsion balance. Per-motor corrected input power increased from 316.49 W in the zero-payload run to 665.20 W in the 10 kg run. Modeled loss increased from 34.60 W to 64.88 W.

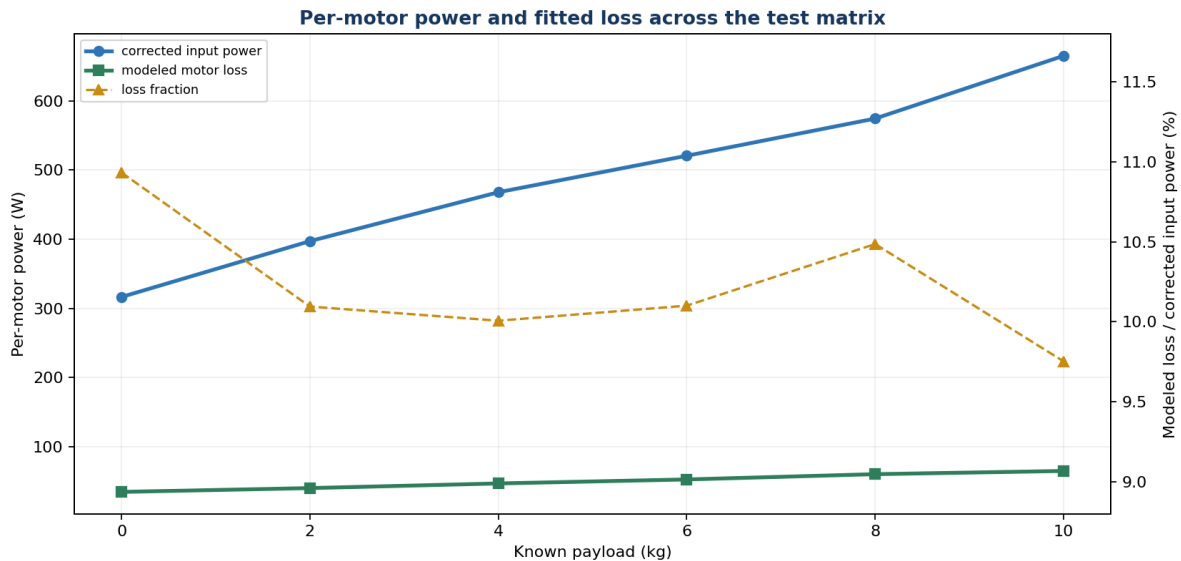


Figure 5. Final per-motor corrected input power, modeled loss and modeled loss fraction across the payload matrix.

The modeled loss fraction remained close to 10% across the matrix, varying from approximately 9.75% to 10.93%. This does not by itself imply that loss is constant: absolute modeled loss nearly doubled, while useful induced power also rose strongly with supported thrust.

9.4 Error Sources and Interpretation

Tare-mass sensitivity: payload is obtained after subtracting 18.75 kg. A small total-mass error therefore becomes a much larger percentage of a 2–4 kg payload.

Zero-load bias: the -0.9441 kg diagnostic result indicates that electronics-baseline subtraction and low-load coefficient transfer require further characterization.

Four-kilogram outlier: the 4 kg trial produced the largest clean-trial APE, 12.92%, despite neighboring 2, 6 and 8 kg trials being much closer. A single flight cannot determine whether this came from hover selection, environment, battery condition or measurement scatter.

Coefficient dependence: R_{copper} , k_{linear} and k_e were obtained from a highly correlated one-dimensional manufacturer operating curve. The combined model fits power well, but individual coefficient interpretation remains calibration-dependent.

Motor averaging: mean PWM and equal current sharing ignore individual rotor trim and motor-to-motor losses.

Repeatability limitation: no repeated trials were available at a given payload, so confidence intervals and between-flight variance cannot be reported.

Overall, the test campaign supports the feasibility of the physics-informed estimator and demonstrates strong total-mass recovery. It does not yet establish a universal sub-5% payload estimator across the full operating range.

10. Future Work and Implementation Roadmap

Implementation priority

The next accuracy improvement should come from measurement discipline and identifiable calibration—not by adding more mutually correlated loss terms. The final model structure should remain fixed while tare mass, baseline power and coefficient uncertainty are characterized with repeated independent data.

10.1 Immediate Calibration Work

Measure the complete aircraft tare mass on a calibrated scale with the same battery, mounts, wiring and test hardware installed.

Characterize motors-disarmed auxiliary current and power at several battery voltages so baseline subtraction does not depend on one short interval.

Repeat the 0, 2, 4, 6, 8 and 10 kg matrix at least three times in randomized order and redo the botched 10 kg flight.

Freeze η_{prop} and the model structure before fitting; fit only R_{copper} , k_{linear} and k_e and retain parameter covariance and sensitivity information.

10.2 Model and Signal-Processing Improvements

Replace the per-flight percentile-only hover rule with an empirically chosen absolute stability threshold, a minimum contiguous duration and additional vertical-speed or attitude-rate checks.

Interpolate RPM separately for C1–C6 and sum per-rotor predictions; where possible, obtain ESC-level current or estimate current sharing from the manufacturer map.

Log motor or ESC temperature so copper-loss temperature dependence can be applied rather than absorbed into one effective resistance.

Use measured ambient temperature and humidity with GPS altitude for a flight-specific density estimate.

Add explicit guards for missing baseline samples, empty stable-hover selections, PWM extrapolation and non-finite intermediate values.

10.3 Validation and Deployment Roadmap

Stage	Focus	Action	Acceptance purpose
1	Calibration dataset	Repeated randomized payload flights; archive aligned CSVs and conditions	Stable coefficient and baseline estimates
2	Held-out validation	Freeze the model and test separate flights/days	Unbiased MAPE, bias and confidence intervals
3	Robustness campaign	Vary battery SOC, temperature, altitude and moderate wind	Operating envelope and correction priorities
4	Fleet maintenance	Standardized periodic no-payload hover or bench check	Compensation for aging, wear and configuration changes
5	Operational integration	Automated warnings, versioned calibration and preserved audit outputs	Repeatable field use

11. Conclusion

Final conclusion

The internship produced a complete physics-informed payload-estimation model, a reproducible BIN-log inference pipeline and an initial field-validation campaign. The final clean loaded trials achieved 5.89% payload MAPE and 1.06% total-mass MAPE; three of four clean 2–8 kg trials were within 5% payload error.

The final model corrects several issues discovered during development: inaccurate data extraction, noisy altitude selection, non-physical polynomial alternatives, misidentified resistance, redundant linear-speed coefficients, an outdated 18.5 kg tare mass and omission of baseline electronics power from the total-power balance. The final coefficients are $R_{\text{copper}} = 0.09667 \, \Omega$, $k_{\text{linear}} = 4.322131 \times 10^{-3} \, \text{W/RPM}$, $k_e = 1.603907 \times 10^{-6} \, \text{W/RPM}^2$ and $\eta_{\text{prop}} = 0.718$.

The field results show that the estimator already recovers supported aircraft mass with low average error, but payload percentage error is amplified because the payload is the residual after subtracting an 18.75 kg tare mass. The 4 kg outlier, negative zero-payload diagnostic and botched 10 kg test prevent a general claim of sub-5% payload accuracy at this stage.

The work therefore concludes with a functioning and technically defensible estimation framework, clear validation evidence and a practical path to improved accuracy through direct tare measurement, repeat tests, better baseline characterization, identifiable coefficient fitting and stronger software guards. These outcomes satisfy the internship objective of developing and validating a telemetry-based payload-estimation method while documenting its assumptions and present limitations.

Appendix A. Manufacturer PWM-to-RPM Lookup

Throttle command	RPM
40%	2455
42%	2605
44%	2754
46%	2900
48%	3047
50%	3184
52%	3325
54%	3456
56%	3589
58%	3716
60%	3841
62%	3959
64%	4080
66%	4194
68%	4307
70%	4415
72%	4519
74%	4627
76%	4723
78%	4821
80%	4912
90%	5142
100%	5442

Source: Hobbywing X6 Plus power-system technical parameter sheet; final pipeline uses linear interpolation between these points.

Appendix B. Code-to-Equation Traceability

Implementation expression	Meaning	Report reference
<code>merged['Power_total'] = Volt × Curr</code>	Electrical input measurement	$P_{\text{total}} = VI$
<code>PWM_avg = mean(C1...C6)</code>	Mean six-motor command	PWM
<code>rho = rho_0 × (...)**exponent</code>	Air density	Equation 7
<code>prop_wash_area = πr^2 – A_lost</code>	Effective disk area	Equation 4
<code>RPM = np.interp(...)</code>	Motor-speed estimate	Equation 8
<code>P_motor_loss = $R I^2 + k_{\text{linear}} \text{RPM} + k_e \text{RPM}^2$</code>	Final identifiable motor-loss model	Equation 6
<code>T_total = ((P_total – P_loss) $\eta \sqrt{2\rho A}$))**(2/3)</code>	Per-rotor thrust	Equation 9
<code>Total_weight = (T/g) × 6</code>	Total supported mass	Equation 10, first term
<code>Weight_Payload = Total_weight – 18.75</code>	Payload mass using corrected tare	Equation 10
<code>P_corrected = P_hover – P_aux</code>	Electronics-baseline power correction	Section 6.4

Appendix C. Final Pipeline Preconditions

14. The log contains GPS, CTUN, RCOU and BAT messages with the required fields.
15. BAT Curr is instantaneous current in amperes; CurrTot is not substituted.
16. GPS Alt is in metres and represents a consistent altitude reference across tests.
17. CTUN SAlt is a valid rangefinder-based height near the ground and during hover.
18. RCOU C1–C6 correspond to the six lift motors in the tested configuration.
19. The tested motor, propeller, ESC and PWM endpoint configuration matches the calibration.
20. The empty-aircraft mass used by the code matches the actual aircraft configuration.
21. The log includes both an SAlt < 0.05 m baseline and a stable-hover interval above 3.5 m.
22. The stable hover is not dominated by strong gusts, large attitude changes, or continuous climb/descent.

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30. Final payload-weight inference pipeline and curve-fitting implementation supplied with this report, revised 30 July 2026.

Document Control

Field	Value
Document	Payload Weight Estimation – End-of-Internship Technical Report
Revision	Final – model, field results and internship completion
Date	31 July 2026
Pending additions	None for internship submission; future validation work is documented in Section 10
Primary code artifact	testing_pipeline_final.py